

Lessons from the Real-Data Track

fracres meets ds002778: implementation, failures, and validations

Working note — fracres project (D. R. Chin)

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Abstract

This note records what was learned while taking fracres from a phantom generator to an instrument that loads real EEG, reconstructs held-out channels, and infers generative mechanisms (ROADMAP “Next 1”, commits [92b473a](#), [d96da46](#), [8abc81e](#)). It covers the three scientific findings, the failures encountered en route and how each was resolved, the validation patterns that proved trustworthy, and the open threads. The failures are recorded deliberately: several were cheap to make and would be cheap to make again.

1 What was built

Three modules over the UC San Diego resting-state Parkinson’s dataset (OpenNeuro [ds002778](#): 15 PD patients $\times$ medication OFF/ON sessions, 16 controls; BioSemi 32-channel, 512 Hz, $\sim$ 200 s runs; 571 MB mirrored locally from the anonymous S3 bucket):

1. **Ingestion** (`fracres.data`): BIDS- and session-aware loading (`ses-off/ses-on/ses-hc`), participants metadata, EXG/status channel dropping, standard-montage geometry, optional re-sampling, JAX hand-off. `mne` is a lazy optional extra.
2. **Forward fit** (`fracres.forward_fit`): average re-reference + z-score, drive the frozen reservoir with a real channel, closed-form ridge reconstruction of held-out channels (temporal split + washout), per-channel held-out correlation/nMSE and DFA- H / spectral- β comparison.
3. **Inverse problem** (`fracres.inverse`): weighted Euclidean objective on the (DFA- H , spectral- β) metric vector; declarative (α, H, λ) grid search with common random numbers, seed-ensembled metrics, length-matched simulation, and an `fbm` drive regime.

2 Scientific findings

1. **The OFF/ON asymmetry (forward fit).** Driven by Fz of `sub-pd3`, the same (α, H) phantom reconstructs seven channels at mean correlation 0.52 OFF medication but only 0.19 ON; the data’s statistics shift between sessions (spectral β up to 1.5–1.8, DFA- H to 1.3–1.45). No single mechanism matches both medication states — the inverse problem is empirically necessary, not merely planned.

2. The fGn regime mismatch, and its resolution. Raw scalp EEG is fBm-like: $\beta \approx 1.8$, DFA- $H \approx 1.3$ (nonstationary). A *contractive* reservoir driven by stationary fGn cannot produce $\beta > 1$ — the first grid sweep bottomed out at objective ≈ 1.58 with every candidate near DFA- $H \approx 0.7$, $\beta \approx 0.4$. The failure was itself the diagnostic: the drive must be integrated. With `drive_kind="fbm"` (z-scored cumulative fGn), the stable reservoir passes the drive’s low-frequency power law through, so *anti-persistent* H reaches the EEG regime exactly: $\beta = 2H + 1$, DFA- $H = H + 1$.

3. The medication contrast (inverse problem). Best-fit mechanism for sub-pd3 Cz (60-point grid, 2 seeds): OFF $\rightarrow (\alpha=0.9, H=0.7, \lambda=1)$, objective 0.068; ON $\rightarrow (\alpha=0.9, H=0.6, \lambda=1)$, objective 0.008 (DFA- H 1.398 vs. data 1.403; β 1.94 vs. 1.94). Dopamine leaves the fractional *order* unchanged and *whitens the effective drive* (H : 0.7 $\rightarrow$ 0.6). Caveats: one subject, α at the grid boundary, H spacing 0.1. A hypothesis, not a result.

3 Failures encountered and how they were resolved

Recorded most-recent-first within each category; each entry names the failure, the signal that exposed it, and the fix.

3.1 Latent code defects

signal_metrics argument misrouting.

The bundle passed `n_segments=8` into `spectral_exponent`’s `f_max` slot, so every bundled β was silently a *full-band* fit ($f_{\max} = 8$ cycles/sample — above Nyquist) — precisely the bias the $f_{\max}$ restriction exists to prevent. The existing test tolerated it because its 0.15 agreement slack absorbed the underestimate. *Fix*: signature changed to `signal_metrics(x, f_max=0.1)`; regression test pins exact equality with `spectral_exponent(x)` and the direction of the band-restriction effect. Estimator recovery improved visibly ($H=0.8$: β 0.555 vs. true 0.6). *Lesson*: positional arguments across similarly-shaped signatures are a standing hazard; bundle functions should forward by keyword, and loose agreement tests hide misrouted arguments.

Suite-order-dependent test failure (global x64).

A new regression test passed standalone but failed in the full suite: other test modules enable `jax_enable_x64` globally at import, and the Davies–Harte fGn differs under float32/float64 — float32 adds a high-frequency noise floor that flattens the spectrum enough to flip an inequality ($\beta_{0.1} = 0.587$ vs. full-band 0.401 in float32; 0.620 vs. 0.622 in float64). *Fix*: the directional assertion now uses a prescribed-spectrum signal (exactly $f^{-0.6}$ below $f=0.1$, flat above, random phases), which is strictly ordered in any precision. *Lesson*: never write order-dependent assertions on stochastically generated signals when global precision state varies; construct the property you want to test directly.

Asserting an invariant in the wrong domain.

My own preprocessing test asserted that average-referenced channels sum to zero *after* per-channel z-scoring — false, because z-scoring rescales channels unequally. *Fix*: assert the zero-sum property with `zscore=False` and the unit-variance property separately. *Lesson*: state which processing stage each invariant belongs to; composing transforms composes their side conditions.

3.2 Wrong assumptions about the dataset

All three of these would have been avoided by inspecting one subject’s files before writing the loader — they were caught by tests against the real mirror instead:

- **Session level.** `ds002778` is not `sub-*/eeg/` but `sub-*/ses-*/eeg/`, and PD subjects have *two* sessions (`ses-off/ses-on`, medication state) — which turned out to be the most scientifically valuable axis of the whole track. `recording_path` now resolves sessions, auto-selecting a single session and requiring an explicit choice when there are several.
- **Channel count.** Assumed BioSemi 64 (from the EXG1–8 discussion in the dataset comments); the cap is 32 scalp channels. Default montage corrected to `biosemi32` (and the mne naming has no underscore — `biosemi_64` fails with a long allowed-values error).
- **Participants schema.** No `Group` column; group comes from the subject-label prefix. The parser now falls back to label inference.

3.3 Test-calibration failures

- **Arbitrary margins.** The inverse-problem recovery test first demanded every non-true grid point score > 0.05 ; the runner-up scored 0.039 — legitimately, since α is the soft direction at finite T . Replaced with *structurally motivated* assertions: the true point scores ≈ 0 (exact replay via common random numbers), wrong- H points lose by ≥ 0.3 (the sharp direction). Test now encodes an identifiability claim, not a magic number.
- **Mental arithmetic in expected values.** A weighted-distance test expected 0.4 where the definition gives $\sqrt{4 \cdot 0.1^2} = 0.2$; the module was right. Weights multiply squared deviations — document the convention at the signature.

3.4 Environment and access

- OpenNeuro needs neither DataLad nor the AWS CLI: the `openneuro.org` S3 bucket allows anonymous `ListBucket` and `GET`, so a 60-line `stdlib` script mirrors a dataset (resumable, size-verified). `mne.export` to EDF additionally needs `edfio`; BDF export is unsupported, so synthetic-test fixtures are EDF.

4 Validation patterns that earned trust

1. **Exact replay (common random numbers).** Sharing model/drive seeds across grid points makes the true mechanism score exactly 0 in a recovery test — a deterministic, epsilon-free identifiability check that doubles as variance reduction in the search itself.
2. **Expressible vs. unpredictable channels.** The forward-fit unit test pairs a lagged/smoothed target (must reconstruct) with an independent-noise channel (must not), so the pipeline is tested at both ends of its operating range.
3. **Opt-in integration tests.** Real-data tests skip unless the local mirror exists (same pattern as the `hpfracc` cross-reference): CI stays data-free, but every structural assumption is verified against the real recordings on a developer machine.

4. **Metrics with physical priors.** Knowing EEG *should* sit at $\beta \in [1, 2]$ turned the inverse problem’s large residual from a disappointing number into a mechanism diagnosis (Section 2, finding 2). Sanity ranges are part of the instrument.

5 Open threads

- Grid refinement: $\alpha \rightarrow 0.95/1.0$ (best-fit is at the boundary), H at 0.05 spacing, more seeds for MLE-tight error bars.
- All-subject $\times$ all-channel OFF/ON sweep — the population-level version of finding 3, and the seed of the phantom-vs-real-EEG study (ROADMAP Next 5).
- Per-node (α, λ) (Next 2) as the mechanism question moves from “which order” to “which distribution of orders”.
- The `fbm` drive regime currently lives in `fracres.inverse`; promoting it to `fracres.drivers` would let every example and the config system use it.
- Data-use obligations for any publication: cite Jackson et al. 2019 (*eNeuro*), Swann et al. 2015 (*Ann. Neurol.*), George et al. 2013 (*NeuroImage Clin.*); email the curators pre-submission; no PD/HC classifier on this n (curators’ explicit request — our use is forward modelling and LRD statistics).